# Ankush Kaundal

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## RESEARCH INTERESTS

Researcher specializing in flood modeling and risk assessment, hydrogeological modeling, and advanced geospatial data analysis combining machine-learning and physics-based numerical models.

## EDUCATION

**Doctor of Philosophy (Ph.D.)** 2020 - Present
Department of Civil Engineering, Indian Institute of Science (IISc), Bengaluru, India.
Thesis: *Parameter Estimation and numerical modelling of groundwater in data-scarce regions.*
Advisor: Prof. Sekhar Muddu
CGPA: 9.0/10

**M.Tech in Civil Engineering (Water Resources)** 2018 - 2020
Indian Institute of Technology (IIT) Guwahati, Assam, India.
Thesis: *Analytical modelling of groundwater and subsurface drainage in sloping lands using homotopy perturbation method.* [thesis link]
Advisor: Prof. Gautam Barua
CGPA: 9.1/10

**B.Tech in Civil Engineering** 2013 - 2017
Jaypee University of Information Technology, Waknaghat, Himachal Pradesh, India.
Thesis : *Estimation of earth work in designing hill roads.*
Advisor: Prof. Ashish Kumar
CGPA: 7.7/10

## INDUSTRY EMPLOYMENTS

**Engineer II**, *AtkinsRéalis* May-2026 - present

- Developing python based post-processing utilities and automation scripts for numerical models and leading UK projects delivery. (Kent Lower Greensand, Wets Midlands Wharf, London Basin, Bromsgrove)
- Providing guidance on using Machine Learning models for flood prediction and numerical model emulation.
- Numerical modelling of the Mole River Basin under climate change and groundwater abstraction scenarios.

**Consultant**, *Smartterra* Jan-2025 - Mar-2026

- Extreme Value Analysis of hydrometeorological extremes to derive design events for flood risk modelling.
- Development of physics-based flood models using HEC-HMS, HEC-RAS for urban flood predictions.
- Development of machine learning models trained on top of physics-based flood models to speed up flood predictions and assess flood risk.
- Deep learning models to classify wetland vegetation in spectrally complex systems using hyperspectral satellite imagery.
- Development of national scale (India) groundwater recharge potential scores by combining multiple GIS datasets at 1km resolution using MCDA-AHP techniques.

**Consultant**, *Adani Greens/Fluxgen Technologies* Jun-2025 - Mar-2026

- Led a team of 6 to build a scalable lake-recharge groundwater model.
- Developed a lake-recharge, transient groundwater model using MODFLOW (packages: RCH, DRN, GHB, EVT, ETS) to assess flow patterns, flow budget and storage changes in aquifer.
- Using MODFLOW outputs to compute water budget of watershed (Storage Change, ET, GW-outflow, Recharge, Runoff)
- Using PEST tool to estimate aquifer parameters like specific yield, hydraulic conductivity and seasonal recharge factors.

## INDUSTRY PROJECTS

- Flood modelling using HEC-RAS/HEC-HMS, flood hazard mapping and risk assessment for a vehicle manufacturing plant in Pantnagar, Uttarakhand. Client suffered huge business losses due to floods in Oct 2021. A flood model was developed to identify the most critical plant units vulnerable to flood risk. A detailed analysis was conducted, and the report was prepared and submitted.
  Client: **TATA Motors** Oct-2022
- Groundwater modeling using MODFLOW for rainwater harvesting and artificial recharge through lakes. Built transient model using GHB, RCH, DRN packages to estimate the aquifer recharge through lakes. Used PEST tool to calibrate the model. Prepared and submitted scientific report to client.
  Client: **TATA Motors** Jun-2022
- Modflow Modelling of Pollutant transport of effluents of paper mill on an irrigated agriculture.
  Client: **Andhra Papers Limited** Nov-2023
- Water supply scheme and water treatment plant design. (Structural & environmental design, DPR preparation)
  Client: **Mastiff Hotels company** Oct-2021

## TECHNICAL SKILLS

**Programming Languages:** Python, R, MATLAB, JavaScript.

**Libraries:** FloPy(MODFLOW), scikit-learn, Pytorch, TensorFlow, Selenium, multiprocessing, joblib, pandas, numpy, rasterio, pyproj, geopandas, SciPy, osgeo(gdal), matplotlib, seaborn.

**Data Science:** Parameter Estimation using Inverse modellling, Monte Carlo Simulations, Time Series Analysis, Random Forests, XGboost, Deep Learning (FCN, CNN, LSTM, U-Net), Deconvolution Techniques (Lasso/Ridge Regression), Optimization algorithms, Parallel Computing.

**IDEs:** Pycharm, VS Code, Spyder, Jupyter Lab, RStudio, Cursor.

**Software Packages:** HEC-RAS, HEC-HMS, MODFLOW 6, MODFLOW-2005 (GMS, ModelMuse), FEFLOW, SWAT, ArcGIS, QGIS, SURFER, AutoCAD, LibreCad.

**Misc:** Google Earth Engine, Linux, GitHub, GitLab, vim, bash, anaconda, Windows, MS Office.

## ACADEMIC PROJECTS

- Modelling sustainable groundwater extraction using numerical models, optimization and sequential parameter estimation to estimate pumping for Bengaluru International Airport (BIAL).
  Designation: Researcher 2024-2025
- Modelling groundwater in cauvery river basin, Ministry of Earth Sciences Project (MoES).
  Designation: Researcher 2021-2023
- A feasibility study on aquifer storage of groundwater by artificial recharge in and around Punjab region,

India.
Designation: Research Fellow 2020

## PUBLICATIONS

1. **Kaundal, A.**, Muddu, S., 2026. *Sequential estimation of aquifer parameters using water table elevations.* Journal of Hydrology [link] [pdf]
2. High-frequency groundwater measurements capture bonus recharge in a humid tropical catchment. Journal of Hydrology (in-review)
3. Kaundal, A. and Muddu, S.: A simplified approach to modelling groundwater dynamics in complex, data-scarce semi-arid basins, EGU General Assembly 2025, Vienna, Austria. [link] [pdf]
4. Approaches to model groundwater in data-scarce basins using multiple data sources. (prepared)

## CONFERENCES/WORKSHOPS

1. **European General Assembly (EGU) 2025**.
   Presented a poster on "A simplified approach to model groundwater dynamics in data-scarce regions"
   Venue : Vienna, Austria — Apr-2025
2. **Indo-French Monsoon School on Hydrology from Space.**
   Venue : IISc Bengaluru, India — Oct-2023
3. **Advanced Data-driven and Artificial Intelligence Tools in Water Resources (AI4Water)**.
   A short course on applying machine learning techniques in hydrology.
   Venue : IIT Roorkee, India — Aug-2023
4. **Workshop on Reference Evapotranspiration and Crop Coefficients.**
   Workshop by Indo French cell, IISc to partition evaporation and transpiration
   Venue : IISc Bengaluru, India — Apr-2023
5. **International Soil Moisture School (IEEE GRSS).**
   Venue : IIT Bombay, India — Mar-2023
6. **Modelling groundwater flow and quality.**
   MIKE powered DHI training course, Kochi India.
   Venue : Kochi, India — Jan-2023
7. **Terrestrial Modelling and High-Performance Scientific Computing.**
   HPSC TerrSys Fall School - Coupled large scale high resolution models
   Venue : University of Bonn, Bonn, Germany — Oct-2022

## TEACHING/COURSE ASSISTANT

- Groundwater Hydrolgy, CE 214, IISc Bengaluru: Taught groundwater modelling using MODFLOW; developed example simulations/assignmnets for class.
- Fluid Mechanics, CE 217, IISc Bengaluru
- Fluid Mechanics, CE 203, IIT Guwahati

## POSITIONS OF RESPONSIBILITY

- Vice President (Finance) of **Entrepreneurship and Innovation cell of IISc Bengaluru**.
- Core Member – **Red Ribbon Club, IIT Guwahati**.
- Core Member – **Media and Publicity, CEC JUIT Waknaghat**.

## ACHIEVEMENTS

- **Mukhyamantri Protsahan Yojna** – Scholarship by Himachal Pradesh Government. 2018
- **Graduate Aptitude Test in Engineering (GATE 2018)**: Secured GATE 99 percentile and achieved Ministry of Human Resources & Development (MHRD) scholarship.